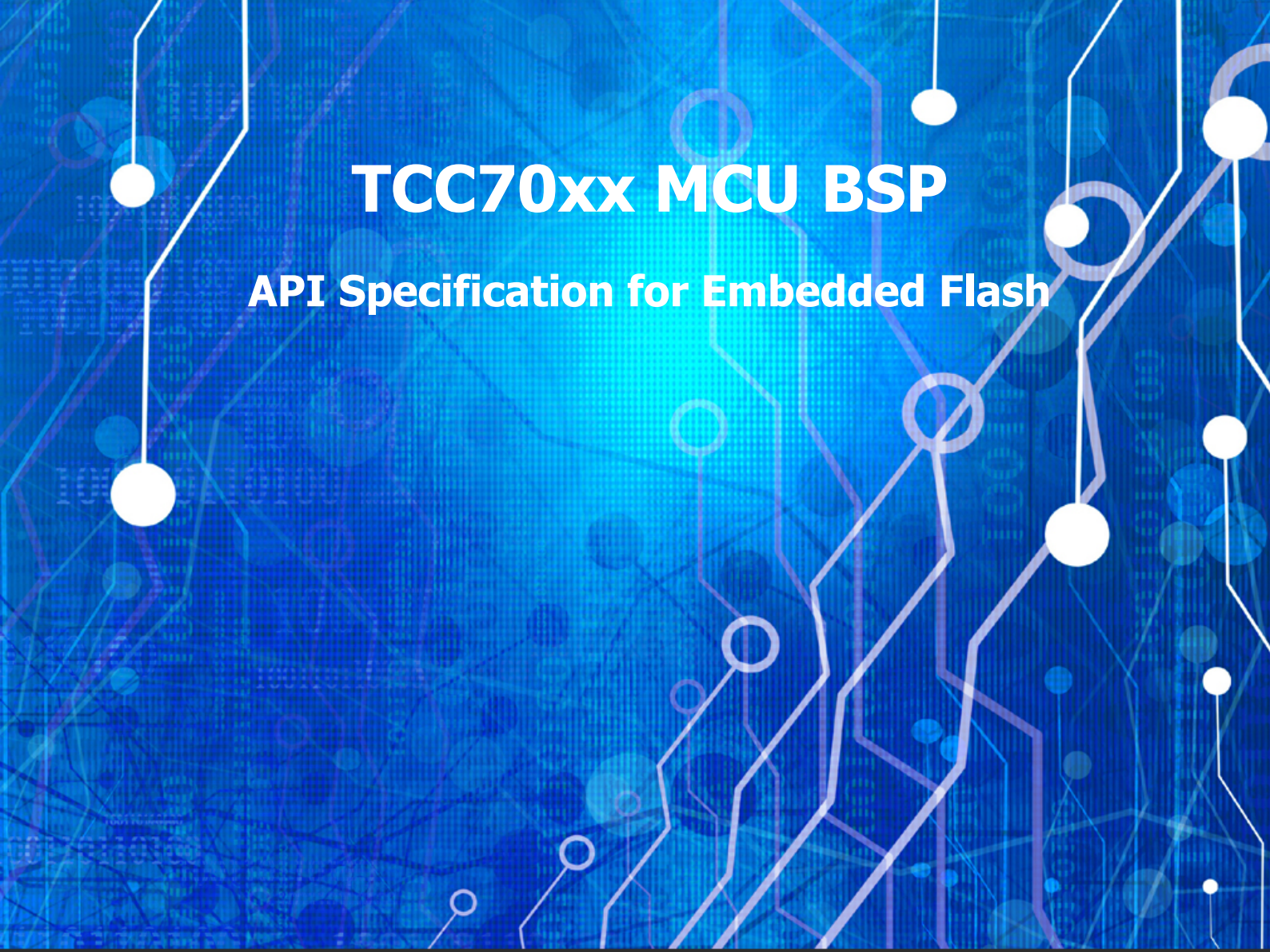




TCC70xx MCU BSP

API Specification for Embedded Flash

Rev. 0.10 [A]

2021-09-03

Preliminary version

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1 INTRODUCTION

This document describes the eFlash device driver of TCC70xx MCU BSP. For more detailed information on eFlash in TCC70xx MCU Sub-system, refer to Chapter 5 and Chapter 6 Program/Data eFlash Memory and Controller in "*TCC70xx Full Specification*"[1].

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

eFlash	Embedded Flash
--------	----------------

3 SOURCE FILES

3.1 Location of Source Files

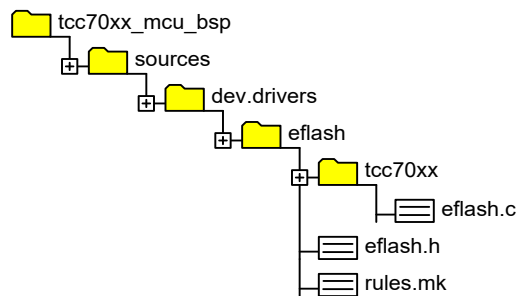


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- eflash.c
 - Driver source of eFlash driver
- eflash.h
 - API header of eFlash driver
- rules.mk
 - Rules used for build

4 CONSTANTS

4.1 EFLASH Type Configuration Flag Values

These constants are used to determine the eFlash type.

There are two types of eFlash controlled by eFlash driver: Program eFlash and Data eFlash.

Declaration

```
typedef enum
{
    EF_PFLASH = 0u,
    EF_DFLASH = 1u,
} EF_type;
```

Description

<i>Value</i>	<i>Description</i>
<i>EF_PFLASH</i>	<i>Program eFlash</i>
<i>EF_DFLASH</i>	<i>Data eFlash</i>

Remark

None

4.2 EFLASH Interface Return Values

The constant values below are the return values of the eFlash driver API.

Declaration

```
enum
{
    EFLASH_RET_OK                = 0u,
    EFLASH_RET_TIMEOUT           = 1u,
    EFLASH_RET_PARAM_OUTOFRANGE = 2u,
    EFLASH_RET_LENGTH_FAIL       = 3u, /* unlikely err report, human err, need to check code */
    EFLASH_RET_DRV_NOT_INIT_YET  = 4u,
    EFLASH_RET_INIT_FAIL         = 5u,
};
```

Description

Value	Description
<i>EFLASH_RET_OK</i>	<i>Success</i>
<i>EFLASH_RET_TIMEOUT</i>	<i>Write, erase operation time out</i>
<i>EFLASH_RET_PARAM_OUTOFRANGE</i>	<i>Input parameter is out of range</i>
<i>EFLASH_RET_LENGTH_FAIL</i>	<i>When size calculation is wrong during write and erase operations</i>
<i>EFLASH_RET_DRV_NOT_INIT_YET</i>	<i>When the interface function is called before the flash driver is initialized</i>
<i>EFLASH_RET_INIT_FAIL</i>	<i>Driver Init Fail</i>

Remark

None

5 STRUCTURES

5.1 EflashRegisterType

The eFlash register is handled by using the following structure.

Declaration

```
typedef struct
{
    volatile uint32    FRWCON;
    volatile uint32    FSHSTAT;
    volatile uint32    reserved[10];
    volatile uint32    TM_LDTCON;           /* offset 0x30 */
    volatile uint32    TM_FSH_CON;
    volatile uint32    TM_ECCCON_1;
    volatile uint32    TM_ECCFADDR_1;
    volatile uint32    TM_DCT0;
    volatile uint32    TM_DCT1;           /* offset 0x44 */
    volatile uint32    reserved1[6];
    volatile uint32    TM_LDT0;           /* offset 0x60 */
    volatile uint32    TM_LDT1;
    volatile uint32    reserved2[38];
    volatile uint32    DCYCRDCON;         /* offset 0x100 */
    volatile uint32    DCYCWRCN;
    volatile uint32    reserved3[62];
    volatile uint32    FSHADDR;          /* offset 0x200 */
    volatile uint32    FSHWDATA_1;
    volatile uint32    FSHWDATA_PRT_1;
    volatile uint32    reserved4;
    volatile uint32    FSHWDATA_2;       /* offset 0x210 */
    volatile uint32    FSHWDATA_3;
    volatile uint32    FSHWDATA_4;
    volatile uint32    FSHWDATA_PRT_2;
} EflashRegisterType;
```

Description

Value	Description
FRWCON	Flash Read/Write Control register
FSHSTAT	Flash Status register
reserved[10]	Not used register area
TM_LDTCON	LDTCON Register for test mode only
TM_FSH_CON	Test Mode Flash Control Register
TM_ECCCON_1	ECC Option Control register
TM_ECCFADDR_1	ECC fail Address
TM_DCT0	DCT0 Register

<i>Value</i>	<i>Description</i>
<i>TM_DCT1</i>	<i>DCT1 Register</i>
<i>reserved1[6]</i>	<i>Not used register area</i>
<i>TM_LDT0</i>	<i>LDT0 Register</i>
<i>TM_LDT1</i>	<i>LDT1 Register</i>
<i>reserved2[38]</i>	<i>Not used register area</i>
<i>DCYCRDCON</i>	<i>Delay Cycle Read Control register</i>
<i>DCYCWRCN</i>	<i>Delay Cycle Write Control register</i>
<i>reserved3[62]</i>	<i>Not used register area</i>
<i>FSHADDR</i>	<i>Flash Address for Program</i>
<i>FSHWDATA_1</i>	<i>Write Data[31:0] for Program</i>
<i>FSHWDATA_PRT_1</i>	<i>Write N bit Data for Program on ECC OFF mode</i>
<i>reserved4</i>	<i>Not used register area</i>
<i>FSHWDATA_2</i>	<i>Write Data[63:32] for Program</i>
<i>FSHWDATA_3</i>	<i>Write Data[95:64] for Program</i>
<i>FSHWDATA_4</i>	<i>Write Data[127:96] for Program</i>
<i>FSHWDATA_PRT_2</i>	<i>Write N bit Data for Program on ECC OFF mode</i>

Remark

None

6 APIs

6.1 EFLASH_Init

Prepares eFlash data buffer.

Declaration

```
uint32 EFLASH_Init
(
    void
);
```

Parameters

None

Return

Value	Description
EFLASH_RET_OK	Success

Remark

None

6.2 EFLASH_FWDN_LowFormat

Erases eFlash entire memory.

Declaration

```
uint32 EFLASH_FWDN_LowFormat  
(  
    EF_type uiEfType  
);
```

Parameters

Value	Description
<i>uiEfType</i>	<i>[in] eFlash type Refer to Chapter 4.1.</i>

Return

Value	Description
<i>EFLASH_RET_OK</i>	<i>Success</i>
<i>EFLASH_RET_TIMEOUT</i>	<i>Write, erase operation time out</i>
<i>EFLASH_RET_PARAM_OUTOFRANGE</i>	<i>Input parameter is out of range</i>
<i>EFLASH_RET_LENGTH_FAIL</i>	<i>When size calculation is wrong during write and erase operations.</i>
<i>EFLASH_RET_DRV_NOT_INIT_YET</i>	<i>When the interface function is called before the flash driver is initialized</i>

Remark

None

6.3 EFLASH_FWDN_Read

Reads data from eFlash.

Declaration

```
uint32 EFLASH_FWDN_Read
(
    uint32    uiAddress,
    uint32    uiLength,
    void *    pBuf,
    EF_type    uiEfType
);
```

Parameters

Value	Description
<i>uiAddress</i>	<i>[in] eFlash start address to be read, Physical address offset value based on eFlash</i>
<i>uiLength</i>	<i>[in] Length of data to be read, byte unit</i>
<i>pBuf</i>	<i>[out] Destination buffer to copy read data</i>
<i>uiEfType</i>	<i>[in] eFlash type. Refer to Chapter 4.1.</i>

Return

Value	Description
<i>EFLASH_RET_OK</i>	<i>Success</i>
<i>EFLASH_RET_TIMEOUT</i>	<i>Write, erase operation time out</i>
<i>EFLASH_RET_PARAM_OUTOFRANGE</i>	<i>Input parameter is out of range</i>
<i>EFLASH_RET_LENGTH_FAIL</i>	<i>When size calculation is wrong during write and erase operations.</i>
<i>EFLASH_RET_DRV_NOT_INIT_YET</i>	<i>When the interface function is called before the flash driver is initialized</i>

Remark

None

6.4 EFLASH_FWDN_Write

Writes data to eFlash.

Declaration

```
uint32 EFLASH_FWDN_Write
(
    uint32    uiAddress,
    uint32    uiLength,
    void *    pBuf,
    EF_type    uiEfType
);
```

Parameters

Value	Description
<i>uiAddress</i>	<i>[in] eFlash start address to be written, Physical address offset value based on eFlash</i>
<i>uiLength</i>	<i>[in] Length of data to be written, byte unit</i>
<i>pBuf</i>	<i>[in] Source buffer to be written</i>
<i>uiEfType</i>	<i>[in] eFlash type. Refer to Chapter 4.1.</i>

Return

Value	Description
<i>EFLASH_RET_OK</i>	<i>Success</i>
<i>EFLASH_RET_TIMEOUT</i>	<i>Write, erase operation time out</i>
<i>EFLASH_RET_PARAM_OUTOFRANGE</i>	<i>Input parameter is out of range</i>
<i>EFLASH_RET_LENGTH_FAIL</i>	<i>When size calculation is wrong during write and erase operations.</i>
<i>EFLASH_RET_DRV_NOT_INIT_YET</i>	<i>When the interface function is called before the flash driver is initialized</i>

Remark

None

7 HOW TO USE EFLASH DRIVER

The usage is simple, so no further explanation is provided in this chapter. If you need an example of use, refer to the code in `app.sample\app.fw.update\fwupdate.c` because eFlash driver is used for FW update.

8 REFERENCES

- [1] TCC70xx Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

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